Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12	*(a)*	(i)		can't be magnesium since gives colour in flame test (1) can't be (strontium) or barium since compound A / the hydroxide is not soluble enough (1)			2	2		2
		(ii)		calcium carbonate accept $CaCO_3$		1		1		1
		(iii)		n(acid) = 3.92×10^{-3} n(compound **C**) = 1.96×10^{-3} (1) M_r(compound **C**) = 56.1 A_r(O) is 16 therefore A_r of metal must be 40.1 (1)			2	2	2	2
	(b)			number of protons = 50 / relative masses of isotopes are 120 and 122 (1) $A_r = \frac{(120 \times 57.9) + (122 \times 42.1)}{100}$ (1) A_r = 120.8 (1) **must** be given to 4 sig figs		3		3	1	

Question				Marking details	Marks available AO1	AO2	AO3	Total	Maths	Prac
	(c)			**Indicative content** • addition of acid to form soluble salt • excess acid to ensure maximum amount of magnesium salt • addition of soluble carbonate • excess to ensure all magnesium ions precipitated • filter and dry precipitate • appropriate equations **5-6 marks** Full description of two stages; good attempt at two equations *The candidate constructs a relevant, coherent and logically structured method including all key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.* **3-4 marks** Full description of one stage; attempt at one equation *The candidate constructs a coherent account including most of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary are generally sound.* **1-2 marks** Gives reactants or partial description of one stage *The candidate attempts to link at least two relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.* **0 marks** *The candidate does not make any attempt or give an answer worthy of credit.*		4	2	6		6
				Question 12 total	**0**	**8**	**6**	**14**	**3**	**11**